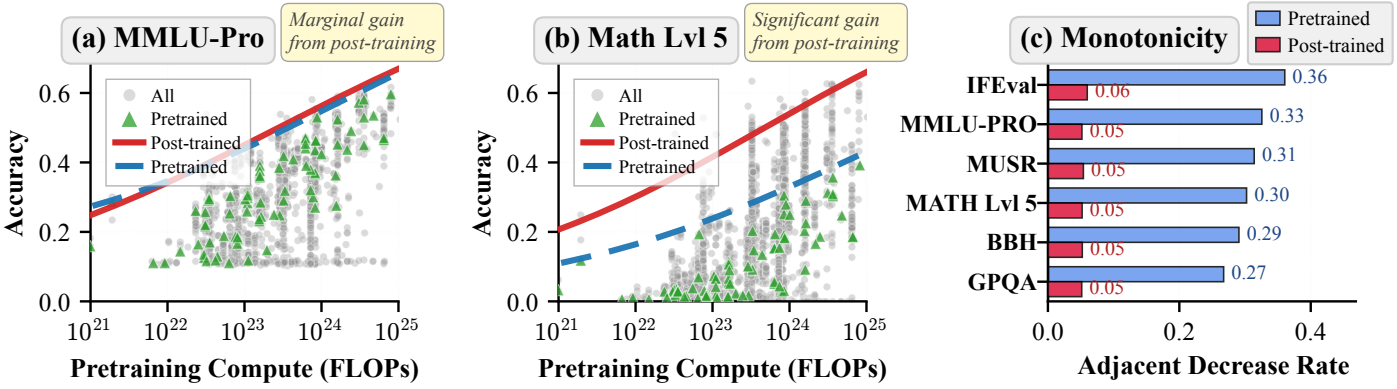
(a) MMLU-Pro
Marginal gain from post-training
All
Pretrained
Post-trained
Pretrained
Accuracy
0.0
0.2
0.4
0.6
10^{21}
10^{22}
10^{23}
10^{24}
10^{25}
Pretraining Compute (FLOPs)
(b) Math Lvl 5
Significant gain from post-training
All
Pretrained
Post-trained
Pretrained
Accuracy
0.0
0.2
0.4
0.6
10^{21}
10^{22}
10^{23}
10^{24}
10^{25}
Pretraining Compute (FLOPs)
(c) Monotonicity
Pretrained
Post-trained
IFEval
0.36
0.06
MMLU-PRO
0.33
0.05
MUSR
0.31
0.05
MATH Lvl 5
0.30
0.05
BBH
0.29
0.05
GPQA
0.27
0.05
0.0
0.2
0.4
Adjacent Decrease Rate